

Secondary storage and memory devices ★ Golden

One college period · 40–45 minutes · Computer Memory

What you will learn

- Explain magnetic, optical and solid-state secondary storage
- Give examples: HDD, USB, SD, CD/DVD/Blu-ray, external drive
- Compare primary memory with secondary storage for a long answer

Secondary memory. Non-volatile storage used to keep programs and data permanently, cheaper and larger than RAM but slower.

Family	How it stores	Examples
Magnetic	Magnetised spots on spinning platters / tape	Hard disk, floppy (old), magnetic tape
Optical	Laser pits on disc	CD, DVD, Blu-ray
Solid state / flash	Electric charge in chips, no moving parts	USB flash, SSD, SD / MicroSD

- Hard disk drive (HDD): large capacity, inside the system unit, moving heads — slower than SSD.
- SSD: flash chips, silent, fast boot, used in modern laptops.
- USB / pen drive / data traveler: portable flash memory.
- SD / MicroSD: phones and cameras.
- Combo drive: CD+DVD reader/writer in one bay.
- External HDD: backup connected by USB.

Primary memory	Secondary memory
On motherboard, used by CPU directly	On disk / USB, used via I/O
Fast, expensive, smaller	Slow, cheap, huge
RAM is volatile	Non-volatile
Measured in GB today	Measured in GB–TB

HOW BIEK ASKS THIS

2026 Section C: “Explain secondary storage devices with examples.” Write magnetic, optical, solid-state with two examples each and one use.

اردو میں سمجھیں

ثانوی میموری مستقل ذخیرہ ہے: ہارڈ ڈسک، یو ایس بی، سی ڈی/ڈی وی ڈی، ایس ڈی کارڈ۔ یہ RAM سے سست لیکن سستی اور بڑی ہے۔

CHECK YOURSELF

1. Why is a USB called solid-state storage?
2. Which is faster for Windows boot: HDD or SSD?

Answers: 1. No moving parts; data in flash chips · 2. SSD

MCQs

1. A USB flash drive is an example of:
 - a. Optical storage
 - b. Magnetic tape
 - c. Solid-state storage
 - d. Cache
2. Blu-ray is a type of:
 - a. Magnetic disk
 - b. Optical disc
 - c. SRAM
 - d. Register

Key: 1=c, 2=b

Short questions (3 marks each)

1. Explain secondary storage devices with examples.
2. Differentiate primary and secondary memory.

Model short answers

1. Secondary storage is permanent: HDD/SSD (magnetic/solid state), CD/DVD/Blu-ray (optical), USB and SD (flash). Used for files, OS install and backup.
2. Primary is fast, CPU-close, costly, RAM volatile. Secondary is slow, cheap, large, non-volatile.

Long questions (board Section C)

1. Explain secondary storage devices with examples. Compare them with primary memory.